Analyzing Text Data

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Agenda

- Text Preprocessing
- Text Vectorization
- Sentiment Analysis

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Let’s begin the discussion by answering a few questions on Text Preprocessing

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Text Preprocessing Quiz

Which of the following tasks are performed during text cleaning?

A. Stemming

B. Lowercasing

C. Removal of Special Characters

D. Adding extra white spaces

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Text Preprocessing

Process of preparing and refining raw text data by removing noise, such as special characters, punctuation, stop words, and irrelevant symbols, to standardize text data

Improves the suitability of the data for analysis and modeling

Stopword Removal	Stemming	Lowercasing	Remove special characters	Strip extra white spaces
Removes common words like “and”, “the”, “is”, etc which often appears frequently and generally do not add ‘contextual value’ to the text	Converts the word into its root form, reducing it to its base or stem, to capture the core meaning	Converts all the words into lower case letters	Removes special characters like “, “”, !, @, etc	Removes extra spaces between the words

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Text Preprocessing - Example

Text Preprocessing is key in natural language processing (NLP). It involves refining raw text for machine understanding and analysis. This process includes eliminating redundant words, reducing words to their root form, standardizing text to lowercase, excluding symbols (!@#$), ensuring clean spaces, and filtering out numerical characters 1234.

Text Cleaning

text preprocess key natur languag process nlp involv refin raw text machin understand analysi process includ elimin redund word reduc word root form standard text lowercas exclud symbol ensur clean space filter numer charact 1234

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Text Preprocessing Quiz

Consider the following three sentences:

"The cat jumped"
"The dog barked"
"The cat chased the dog"

What is the number of dimensions in the vector representation after applying the bag-of-words (BoW) model to all these sentences?

NOTE: Stopwords are not to be removed from the sentences.

A 6

B 9

C 12

D 15

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Bag of Words (BoW) model

Represents text by **counting the frequency of unique words** in a document **without considering the order or structure** of the words

Creates a "bag" (or set) of words in a text corpus, ignoring grammar and word order

Example:

Sentence	the	cat	jumped	dog	barked	chased
The cat jumped	1	1	1	0	0	0
The dog barked	1	0	0	1	1	0
The cat chased the dog	1	1	0	1	0	1

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Text Preprocessing Quiz

Great Learning
POWER AHEAD

Words with high frequency are always considered to be stop words.

A. True

B. False

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Stopwords

Words in any language which do not add much meaning to a sentence.

Can safely be ignored without sacrificing the meaning of the sentence.

Example:

Consider you are analyzing a collection of documents related to cooking recipes.

The word "salt" might appear very frequently across these documents due to its significance in cooking.

However, "salt" will not be considered a stop word in this context because it carries essential meaning within the domain.

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Text Preprocessing Quiz

The main characteristic of a lexicon-based approach in sentiment analysis is that it utilizes a predefined set of words with assigned sentiment scores

A. True

B. False

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Sentiment Analysis

The process of analyzing a piece of text and categorizing it based on the context and emotions expressed within the text

In general, categorization is done as positive, negative, or neutral

Used extensively to analyze end-user feedback, gain insights on the sentiment of the user towards a product/service, and identify areas of improvement

Example:

"The movie was fun, brisk, and imaginative" => **Positive**

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Sentiment Analysis - Lexicon-based Approach

Uses predefined dictionaries (or lexicons) containing words and their associated sentiment labels

Each word in new, unseen text is compared against the lexicons to determine its sentiment label, and then the values are combined to get the prediction

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Happy Learning !

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